A Framework for Emergent Time from Horn-Filling Dynamics

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Abstract

We propose a framework in which spacetime is a growing simplicial set and time is the process of filling its inner horns. Two local, non-deleting moves drive the growth of spacetime: a "sprout" attaches a fresh directed edge ending at a new vertex, and a "fill" completes an unfilled inner horn with a composite edge and its witnessing cell. Each move is a weak homotopy equivalence of the underlying space, so the homotopy type is conserved at every step. Since the geometric realization of the simplicial set forgets edge orientation, this conserved type cannot see any arrow of time; all physical content lives in the directed structure that realization forgets. The completed limit of the filling process is a quasicategory, in which every composition question has an answer while edges remain directed. Each move is an event that consumes cells created by earlier events, the dependency of moves forms a partial order, and a reference frame is a choice of linearization of this order. The entropy of an event is the logarithm of the number of linear extensions of its causal past. This quantity measures how much of the past could have happened in a different order, and grows monotonically, which we prove; a toy implementation provides first numerical support. The arrow of time appears as a broken symmetry. The universe carries an arrow exactly when it is not isomorphic to its own edge-reversal, and pairing the universe with its mirror image restores the symmetry globally, in the spirit of CPT-symmetric cosmologies. We formulate a maximum-entropy inference problem whose solution, conjecturally of Gibbs form, would determine the rule from locality and a fixed update budget alone. We close with the open problems, foremost the recovery of the Lorentz group and of an area law for horizon entropy.

1 Introduction

Discrete approaches to quantum gravity, from Regge calculus [13] to causal dynamical triangulations [3], replace smooth spacetime with combinatorial data: simplices, gluings, and the lengths assigned to them. In Regge calculus the combinatorial scaffold is fixed with only edge lengths allowed to vary. The vertices, simplices, and gluings of the triangulation never change, so no new combinatorial structure is ever introduced. That is enough for classical gravity, where one solves the field equations on a triangulation that is given in advance. The discomfort begins one step later, when Feynman's path integral is applied to the evolution of the universe and the sum runs over all histories. Summing over Regge geometries means integrating over edge lengths, a continuous space in which many assignments describe the same geometry and no canonical measure presents itself. On the other hand, dynamical triangulations [7, 1, 2] escape this by inverting Regge's choice. The lengths are frozen and the combinatorics is set free, so a geometry becomes a gluing pattern of simplices whose shapes are fixed once and for all, and summing over geometries becomes counting gluings.

In both cases the geometry is carried by the gluing and not by any ambient space. Each simplex is flat inside, so curvature can only sit where simplices meet, and it is measured by how far the angles around such a meeting place fall short of a full turn. Figure **??** shows the three cases for triangles glued around a vertex.

Because the combinatorics is now free, the number of simplices differs from history to history, and within a single history the number of simplices in a slice changes from one slice to the next.

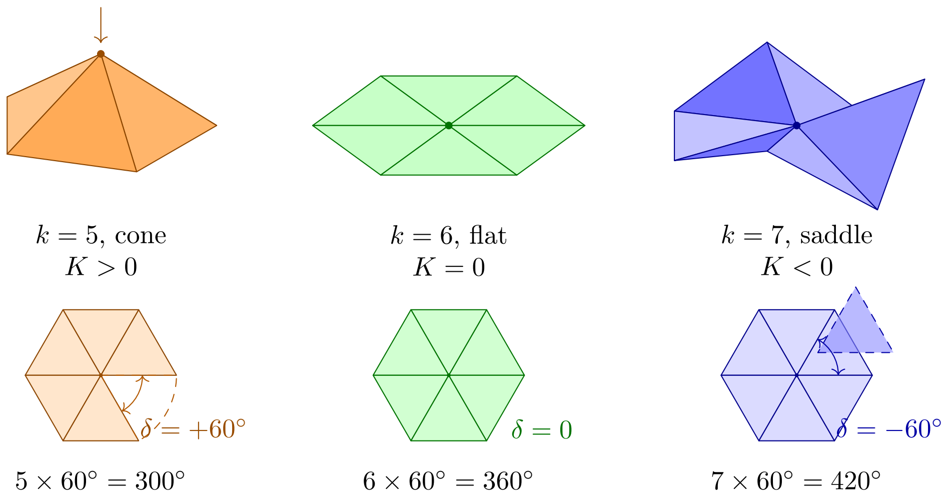


Figure 1: Curvature in Regge calculus lives at the vertices, not on the interiors of the simplices. Gluing k flat unit equilateral triangles around a single point makes the angles there sum to $k \cdot 60°$, and the deficit angle $\delta = 2\pi - \sum_i \theta_i$ measures the failure to close up in the plane. Top row shows the resulting surface in $\mathbb{R}^3$, bottom row shows the same triangles unfolded flat, where δ is directly visible. For $k = 5$ the angles fall short, the gap closes only by lifting the vertex out of the plane, and the cone carries positive curvature concentrated at a point. For $k = 6$ the angles close exactly and the neighbourhood is isometric to a patch of the Euclidean plane. For $k = 7$ the seventh triangle (dashed) overlaps the first, the excess forces the surface to buckle into a saddle, and the curvature is negative. Every triangle is flat throughout, so all the geometry is carried by the combinatorics of how they are glued.

New geometric measure appears from nowhere. Classical general relativity is no better off on this point, since it has no conservation law for spatial volume either, and in a closed universe the Hamiltonian constraint balances the total at zero [4, 8] however much volume appears. The complaint is therefore not that volume grows, but that the growth answers to nothing the theory conserves. A cosmological constant term only assigns a cost to four-volume within the ensemble, and where the total is held fixed instead, as in simulations and in unimodular gravity [15, 10], the constraint is imposed by hand rather than descending from a symmetry.

This paper takes the discomfort seriously and asks what a theory looks like if one demands, from the start, that

1. Nothing is ever destroyed. Evolution proceeds only by forking and completion moves, never by deletion, so that topological invariants are conserved by construction.

2. Time is not a dimension of the fundamental structure but an emergent bookkeeping of an irreversible forking-filling process.

3. Entropy is attached to frame-invariant causal data, not to spatial slices, so that the theory never needs to say what all of space looks like at a single moment.

4. The rule that chooses which move happens next is derived from an inference principle rather than postulated.

In our framework the universe begins as a single vertex, which we call the big bang, and grows by two kinds of local moves:

1. A **sprout** attaches a fresh directed edge ending at a new vertex, adding new space.

2. A **fill** completes an unfilled inner horn, attaching a composite edge together with the higher simplex that witnesses the composition.

Two kinds of direction appear here. Every edge of the simplicial set carries an orientation, and that orientation is what decides which horns count as inner and what composition means. Separately, every move is an event, and the events are ordered by which of them created the simplices the others apply to. We call the first the **orientation** and the second the **causal order**. No move ever deletes anything, so growth always runs forward in the causal order, while orientation is a property of the finished simplices and can be reversed on paper.

At the big bang two edges sprout from the single vertex with opposite orientation, and this is what produces the two branches. The mirror branch is not an independent run. By definition it is the edge-reversal of the branch, carrying one mirrored event for every event, which is what makes the pair a mirror image of itself. The mirroring is causally inert, since no move in one branch ever applies to a simplex created in the other, so the causal past of any event lies wholly inside its own branch. Each branch grows forward in its own causal order while pointing the opposite way in orientation, so neither branch runs backward in time in any absolute sense. Each is backward only as seen from the other, and there is no global causal order in which either of them is reversed. This is a combinatorial counterpart of CPT-symmetric cosmologies, where a universe and an antiuniverse emerge together from a single point [6, 5].

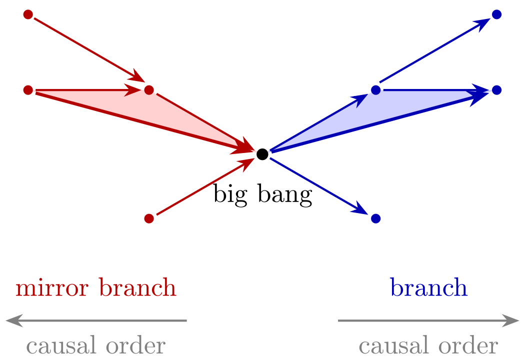


Figure 2: The total object. A branch on the right and its mirror image on the left, glued at the single vertex where both were born. Corresponding edges carry opposite orientation, so the pair is isomorphic to its own edge-reversal. The shaded region on each side is a filled inner horn, drawn with its composite edge in bold, and the fill is mirrored along with everything else. The grey arrows are the causal order, which runs away from the shared vertex on both sides, so each branch is backward only as seen from the other, and never in the order in which its own simplices were built.

At any moment the state of the universe is a finite simplicial set with directed edges, together with the history of how it was built. The limit of the filling process in one branch is a quasicategory, in which every composition has been formed while the orientations remain. The classical, zero-information limit is the nerve of an ordinary category, the simplicial set whose n-simplices are the chains of n composable morphisms [14, 11], and in it every filler is unique. The physical content of a completed universe is precisely how far it is from being a nerve.

The pair of branches is meant to carry no information at all. Two contractible halves glued at a single vertex are again contractible, so every homotopy invariant of the pair is that of a point, and the pair is isomorphic to its own edge-reversal, so it carries no arrow either. Whatever is physical therefore sits inside a single branch, in the orientations and in the history of how its simplices were built.

2 Preliminaries and related work

3 The framework

3.1 The object

At any moment, the universe is a finite simplicial set with directed edges. It begins as a single vertex Δ^0, the big bang, and it grows from there.

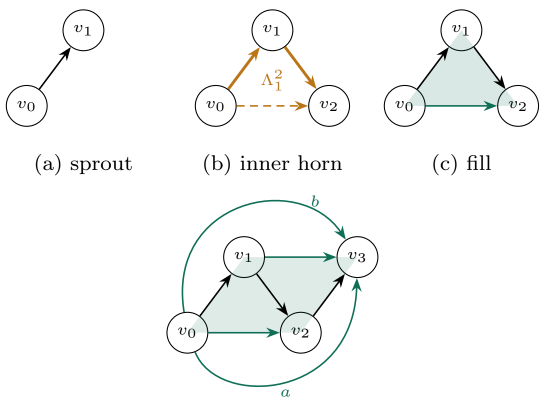
The object is pre-geometric. It is not spacetime, and it is not a representation of a spacetime lying elsewhere. It carries no metric, no dimension, and no manifold structure. Everything geometric has to be recovered from it. Statements about space are made in the framework's own combinatorial terms.

3.1.1 Growing the Simplicial Set

Definition 3.1. The simplicial set grows by two local, non-deleting **moves**, which are pushouts along the following two monomorphisms of finite simplicial sets:

1. **Sprout:** $\Delta^0 \hookrightarrow \Delta^1$. A sprout can happen at any vertex v of the current state. Performing it attaches a fresh directed edge from v to a new vertex.

2. **Fill:** $\Lambda^n_k \hookrightarrow \Delta^n$ for $0 < k < n$. A fill can happen at any unfilled inner horn. Performing it attaches the missing face and interior. Each horn may be filled at most once.



(a) sprout (b) inner horn (c) fill

(d) two composites

Figure 3: The two moves and what each of them adds. In (a) a vertex sprouts a fresh directed edge to a new vertex. After a second sprout, (b) shows the composable pair $v_0 \to v_1 \to v_2$ whose composite is absent, an unfilled inner horn Λ^2_1 (amber, missing edge dashed). In (c) the fill supplies the composite $v_0 \to v_2$ (green) together with the 2-simplex witnessing it (shaded). Panel (d) shows the state after a third sprout and both first-tier fills. Two distinct horns now call for a composite from v_0 to v_3, one built from $v_0 \to v_1$ with $v_1 \to v_3$ and one from $v_0 \to v_2$ with $v_2 \to v_3$. Each fill mints its own composite, so a and b are parallel edges rather than a single shared one. Only once both are present does the inner horn Λ^3_1 exist, since its three faces already carry all six edges of Δ^3; filling it therefore adds a 2-simplex and a 3-simplex but no edge. Only inner horns are ever filled, so edge direction is never inverted and the limit object is a quasicategory.

Remark 3.2. The initial object is a single vertex Δ^0 (the big-bang event). At any moment the current state is a finite simplicial set with directed edges, and the next move is chosen from the set of available moves, which is the disjoint union of all sprout sites and all unfilled inner horns.

Remark 3.3. Every vertex is created by exactly one sprout, which supplies its first incoming edge. Every other incoming edge is a composite supplied by a fill. So sprout edges and composite edges never mix. A sprout creates a fresh vertex and attaches a directed edge to it. The composite joins two vertices that already exist.

Remark 3.4. Sprouting is a Whitehead elementary expansion and inner-horn filling is an anodyne extension. Both operations preserve homotopy equivalences, so the homotopy type of the simplicial set is conserved by construction. In our case, the homotopy type remains that of a point.

Remark 3.5. Only two moves ever add an edge. A sprout adds one, and a fill in dimension two adds one. Fills in dimension three and above add no edges at all, only simplices that say existing paths agree. So every chain is built out of edges, and nothing above dimension one can be part of a chain. Whatever geometric and metric structure this model has is carried by the vertices, the edges, and the causal order between events. Treat the edges as a multigraph. If two fills each produce a composite, keep both as separate edges. The number of parallel edges is a record of how many fills happened there. The order says which events can affect which. The edge counts say how much happened. Both are needed, and neither replaces the other.

3.2 Events, runs, and the causal order

Growth happens one move at a time and never ends. The universe at any moment is a finite stage of an ongoing history, and each event is a move applied at a particular spot. Since no move deletes anything, an event can only apply to simplices that already exist. Event b therefore depends on event a when the spot at which b is applied contains a simplex created by a. The order on events is generated by this dependency, and it is the causal order. A reference frame is a linearization of this order.

Definition 3.6. A **history** is a partially ordered set of events, possibly infinite, in which each event is a sprout or a fill applied at a specified spot of the simplicial set built by the events strictly below it, subject to three conditions: every event has only finitely many events below it, there is a least event called the big bang which creates the initial vertex Δ^0, and the order is generated by creation-before-use.

Definition 3.7. A **run** is a finite initial segment of a history, a finite downset of its events.

At any moment, a run is a partially ordered list of events representing a simplicial set, and the history is the poset that all such stages approximate. Formally, write a run as a pair $(E, \leq)$, where $E = \{e_1, \ldots, e_n\}$ is the set of events and $\leq$ is the causal order on them.

Example 3.8. Start from the single vertex v_0. Event s_1 sprouts an edge $v_0 \to v_1$, event s_2 sprouts an edge $v_1 \to v_2$, and event s_3 sprouts a second edge $v_0 \to v_3$ from the starting vertex. Event f fills the horn $v_0 \to v_1 \to v_2$, supplying the composite $v_0 \to v_2$ and its witnessing 2-simplex. Event s_4 sprouts an edge $v_2 \to v_4$, and event g fills the horn $v_0 \to v_2 \to v_4$, whose first edge is the composite from f. The causal order is generated by $s_1 < s_2 < f < g$ and $s_2 < s_4 < g$, while s_3 shares only the initial vertex with the others and stays incomparable to all of them. The causal past of g is $J^-(g) = \{s_1, s_2, f, s_4, g\}$, in which f and s_4 are incomparable, so $\mathrm{Ext}(J^-(g)) = 2$ and $S(g) = \log 2$. Note that the state at g does not contain v_3, since s_3 is not in the causal past of g.

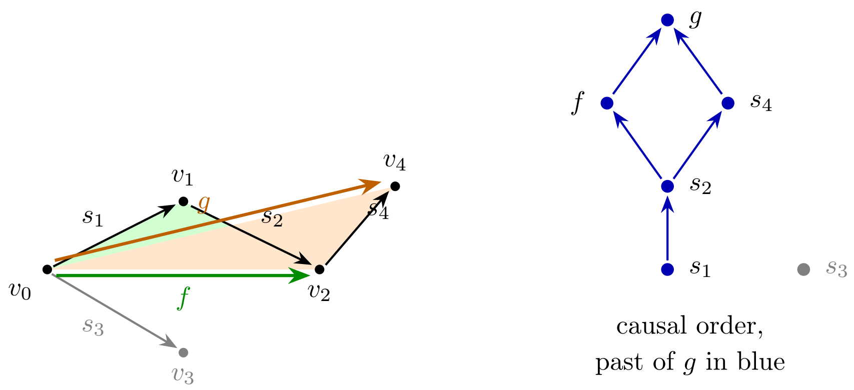


Figure 4: Example 3.8. Left: the simplicial set after the six events, with the composite edges supplied by the fills f (green) and g (orange) and their witnessing 2-simplices shaded to match. Right: the causal order as a Hasse diagram. The events f and s_4 are incomparable, which is what gives $J^-(g)$ its two linearizations. The event s_3 and the vertex v_3 it created (grey) lie outside the causal past of g, so neither enters the state at g.

Definition 3.9. Let e be an event of a history, with causal past $J^-(e) = \{a : a \leq e\}$. The causal past is finite by the definition of a history. The **state at** e is the colimit of the diagram of rule applications in $J^-(e)$.

The colimit is used here so that ordering of $J^-(e)$ has no consequence on the resulting state. Whether the diagram is well-defined, and whether it agrees with what any sequential build produces, is the content of Proposition 3.10.

Proposition 3.10. *The causal order on the events of a history is a partial order, and for every event e the colimit in Definition 3.9 exists and agrees with the state produced by any sequential build along a linearization of $J^-(e)$.*

Proof. Write $(E, \leq)$ for the history and X_D for the colimit of the diagram of rule applications over a finite downset $D \subseteq E$, which exists whenever the diagram is defined since sSet is cocomplete. A move application at b comes with a match $m_b : L \to X_{\{a:a<b\}}$, so every simplex in the image of m_b is created by an event strictly below b. Hence $a \prec b$ implies $a < b$, so dependency is a subrelation of a partial order and therefore acyclic. Its reflexive-transitive closure is then a partial order, which is the causal order.

For the second claim, define X_D by recursion on the size of the downset D. If $D = \varnothing$ then $X_D = \Delta^0$. Otherwise pick a maximal element b of D, set $D' = D \setminus \{b\}$, and define X_D as the pushout of the rule $L \hookrightarrow R$ of b along $m_b : L \to X_{D'}$. The match exists by the admissibility condition above, since every simplex in the image of m_b is created by an event of D'. The causal past $J^-(e)$ is finite, so the recursion terminates and defines the diagram on all downsets of $J^-(e)$, and $X_{J^-(e)}$ is its colimit.

For the third claim, fix a linearization of $J^-(e)$ and consider the sequential build along it. Take two events a and b that are adjacent in the linearization and incomparable in the causal order, applied to a state X. Since a and b are incomparable, neither match uses a simplex created by the other, so both matches factor through X. Pushouts of monomorphisms in sSet are monomorphisms, so the image of m_b persists in $X \cup_{L_a} R_a$ and b can be applied after a. Both orders compute the colimit of the finite diagram formed by X, the two matches, and the two rules, so the two resulting states are canonically isomorphic. This is the local Church–Rosser property for non-deleting double-pushout rewriting [9]. Any two linearizations of a finite poset are connected by a sequence of adjacent transpositions of incomparable elements, so all sequential builds of $J^-(e)$ agree. Finally, a linearization of $J^-(e)$ is a chain containing every object of the poset diagram, so its inclusion into the poset diagram is a final functor, and the iterated pushout along the chain computes the same colimit. Hence every sequential build produces the state of Definition 3.9. □

3.3 Axioms

With the object, the moves, and the causal order in hand, the assumptions of the framework can be stated in the vocabulary just built.

Axiom 1. The fundamental object is the history, an unending partially ordered set of non-deleting moves with the big bang as least event. The state of the universe at an event is the colimit of the diagram of rule applications in its causal past, so it is determined by the history rather than being independent data.

Axiom 2. Evolution proceeds by two local, non-deleting moves: **sprout** and **fill**. Only inner horns may be filled.

Axiom 3. The rate of moves is fixed. The constant c is the exchange rate between translation and internal filling.

Remark 3.11. A rate requires a clock, and time in this framework is emergent, a frame being only a linearization of the causal order. Read as a universal number of moves per tick, the axiom would reinstate a preferred linearization. The intended reading is the energetic one of Margolus and Levitin [12], in which the budget bounds distinguishable changes per unit energy rather than per unit of any global time, and it awaits a notion of energy internal to the framework. The axiom is also silent on whether the budget is one number for the whole universe or one number per region. Locality favours the latter, and the two readings give different cosmologies.

Axiom 4 (Inference)**.** The probability distribution over available moves is the maximum-entropy distribution consistent with Axioms 2–3 and locality (Section **??**). No further dynamical input is allowed.

4 Results

5 Discussion

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